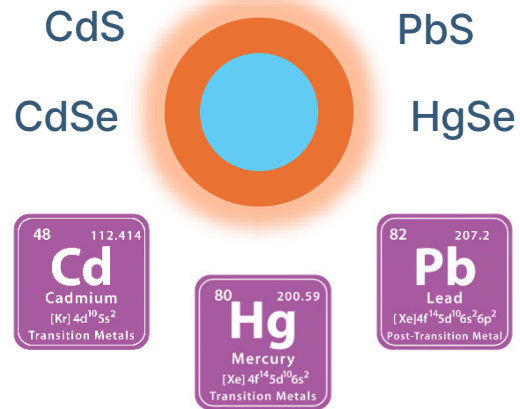


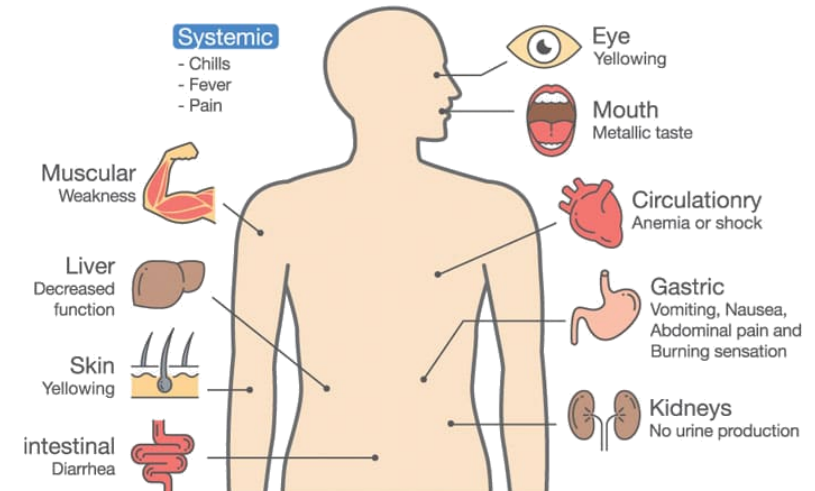
Research Experience: Nanomedicine

Heavy Metal based QD



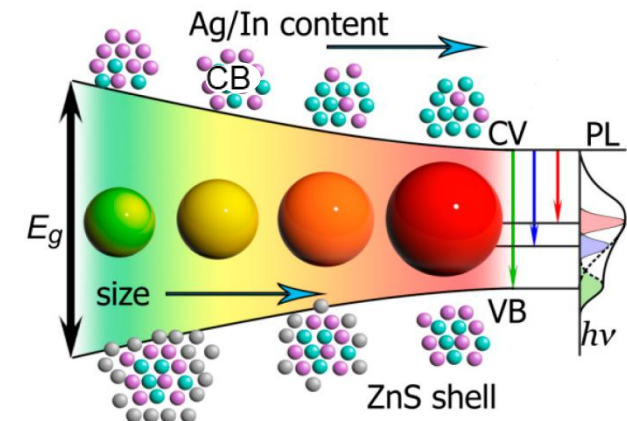
Bio-Application

Causes Biotoxicity



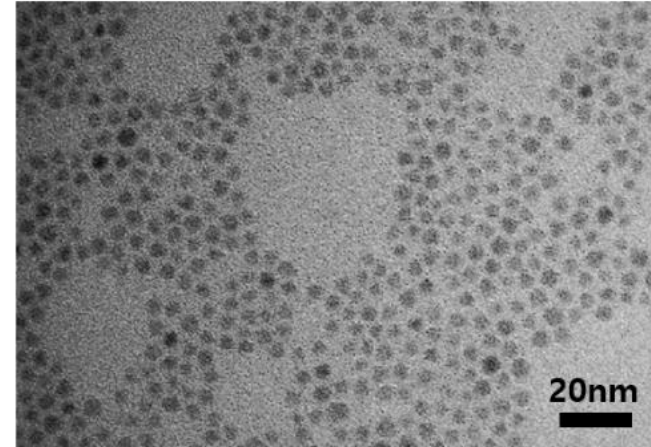
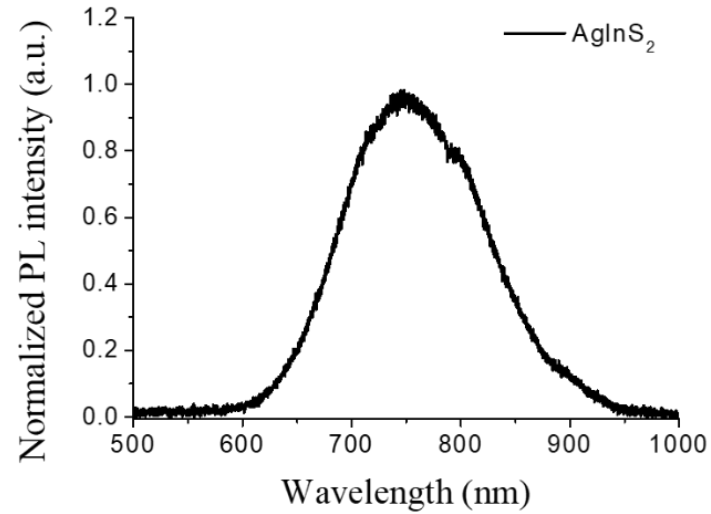
AgInS₂ Quantum Dot

- Heavy metal free non-toxic nanocrystals
- Bio-application for biomarker detecting and therapy
- A large Stokes shift corresponding to the radiative transitions
- Facile tuning of AgInS₂ QD's emission wavelength by controlling Ag/In ratio or coating ZnS shell



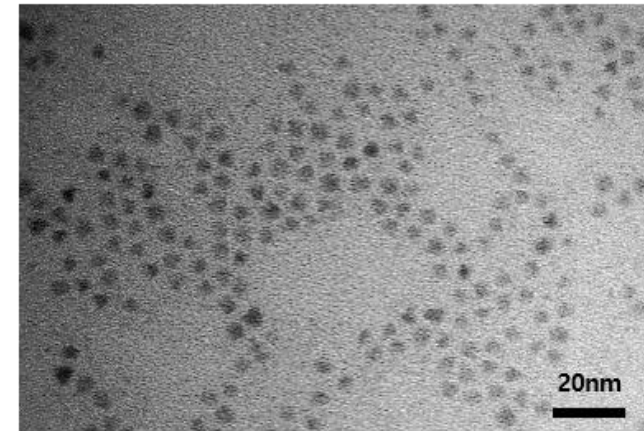
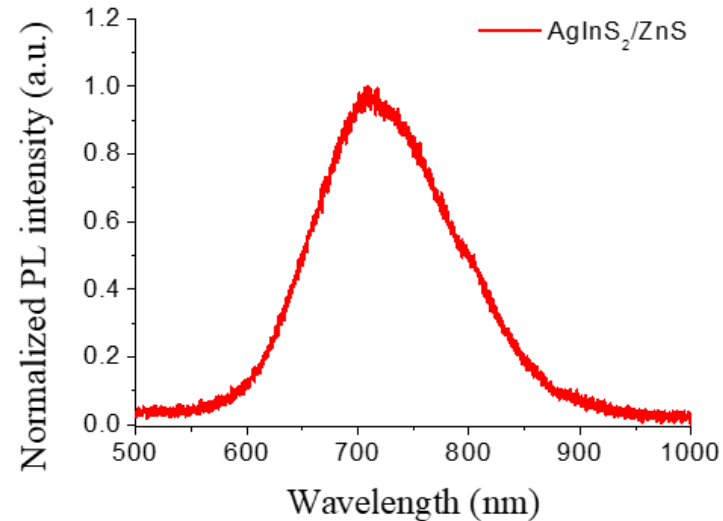
Research Experience: Nanomedicine

AgInS₂ Core



Average size : 3.58 ± 0.36 nm

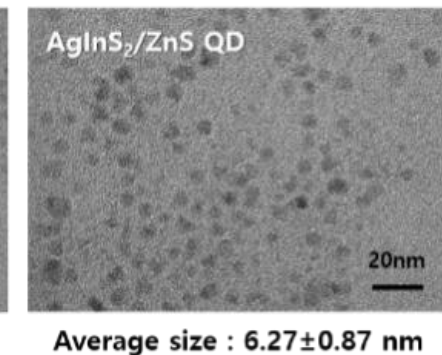
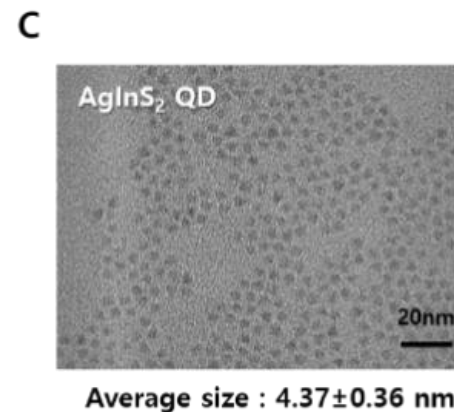
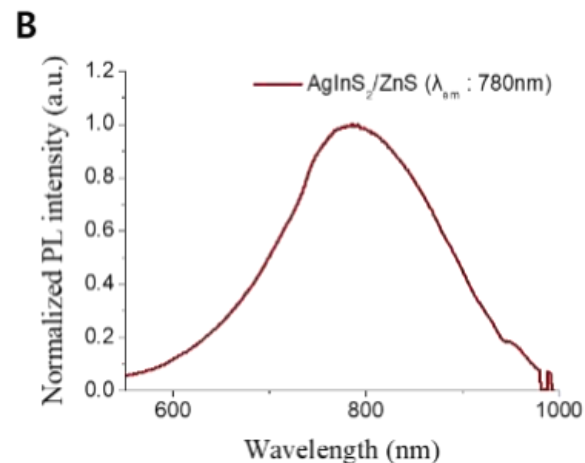
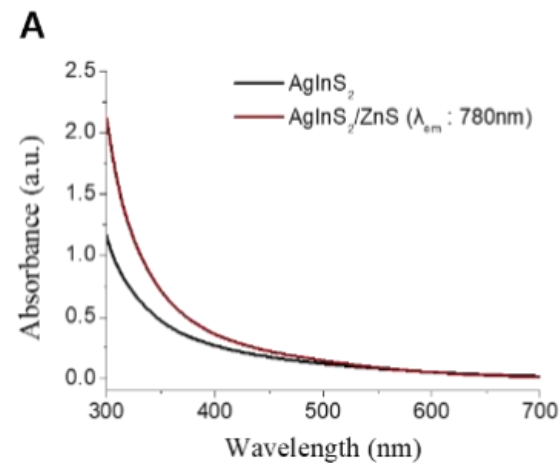
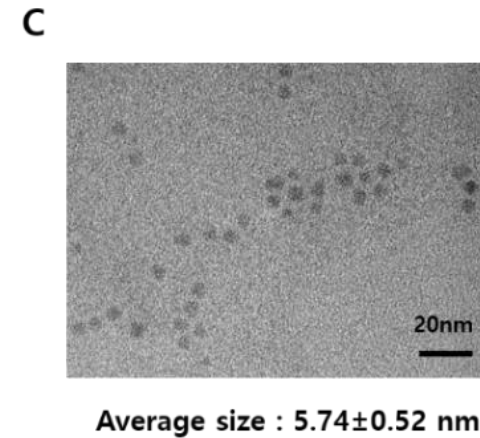
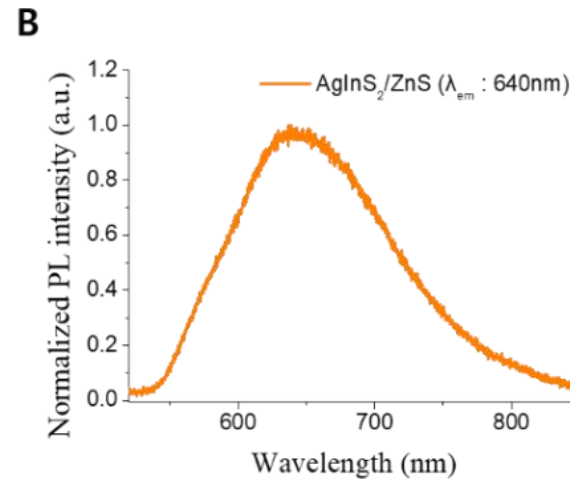
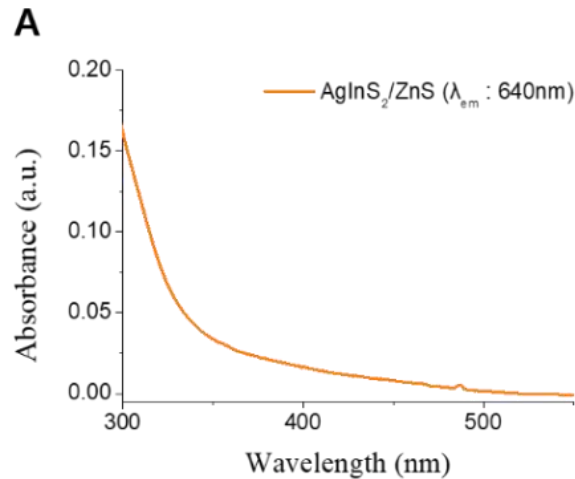
AgInS₂/ZnS Core/Shell



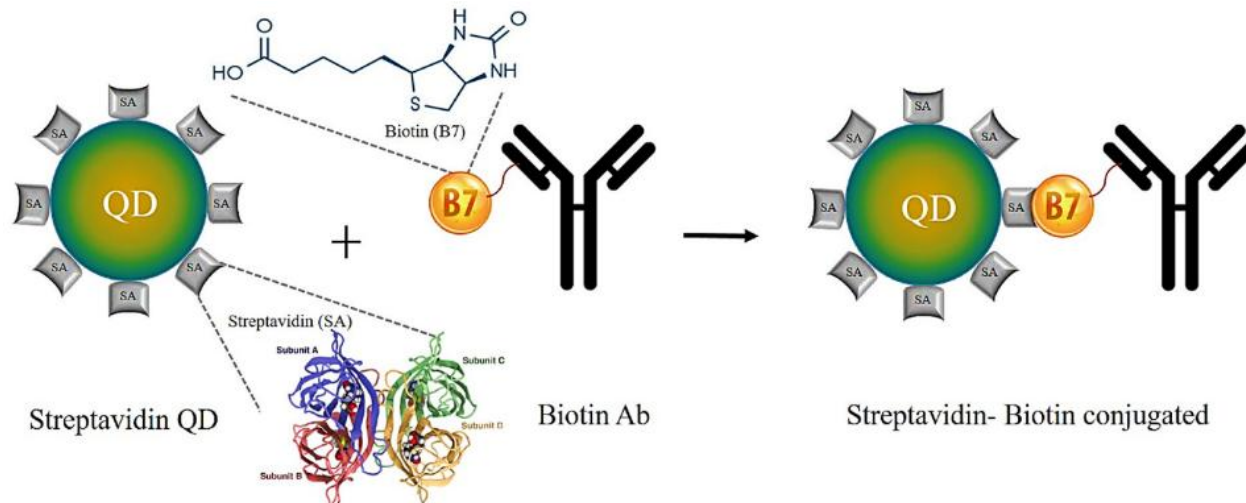
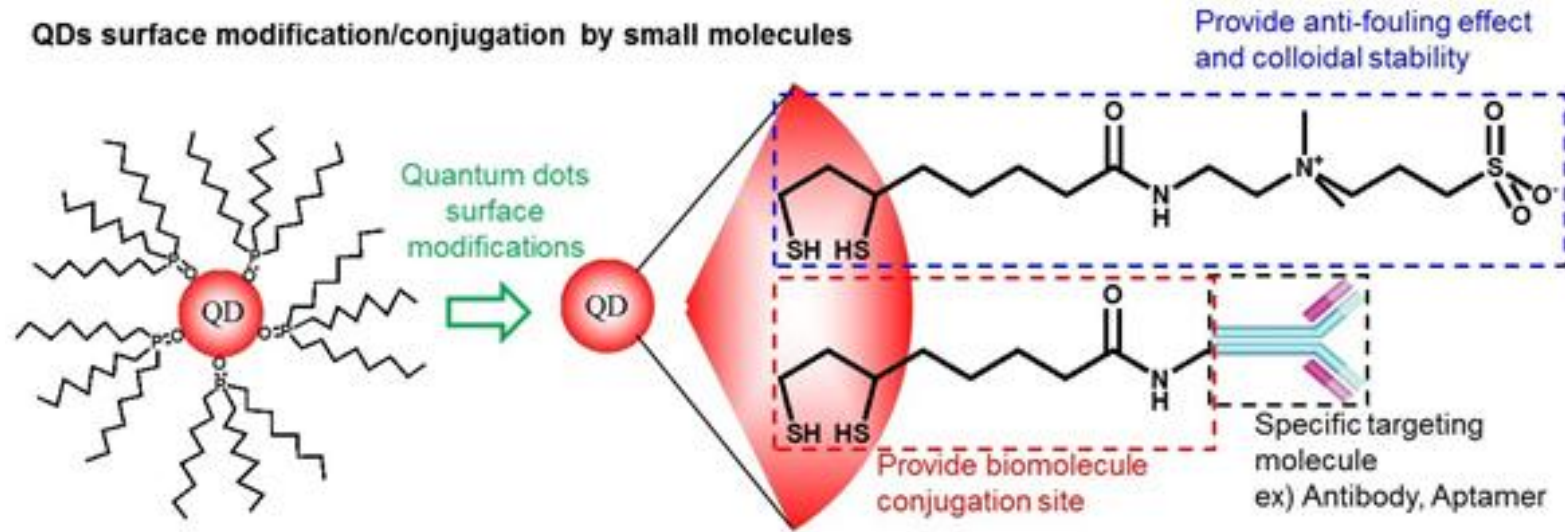
Average size : 4.62 ± 0.40 nm

Research Experience: Nanomedicine

Adjusting the fluorescence wavelength of quantum dots

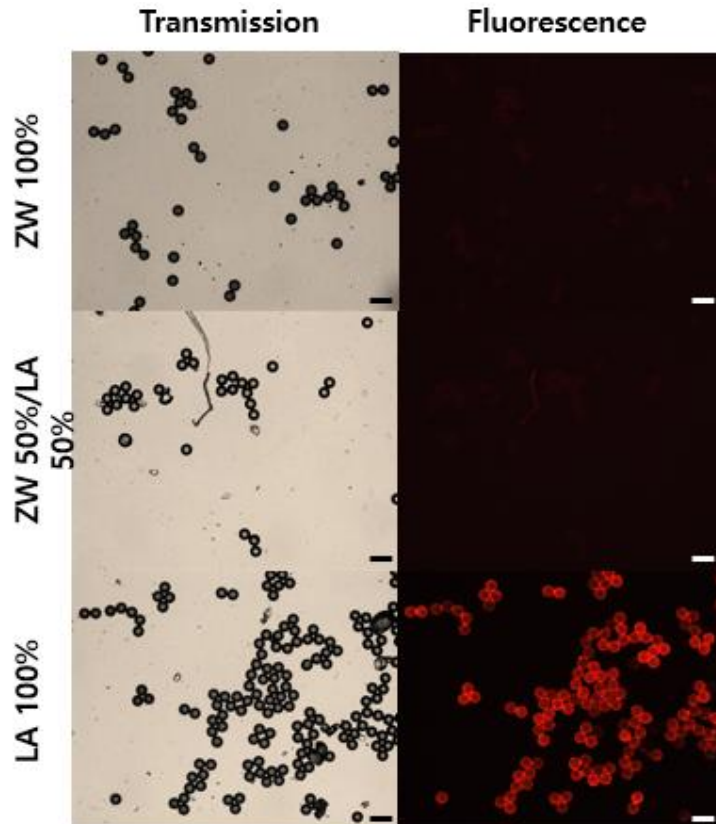


Research Experience: Nanomedicine



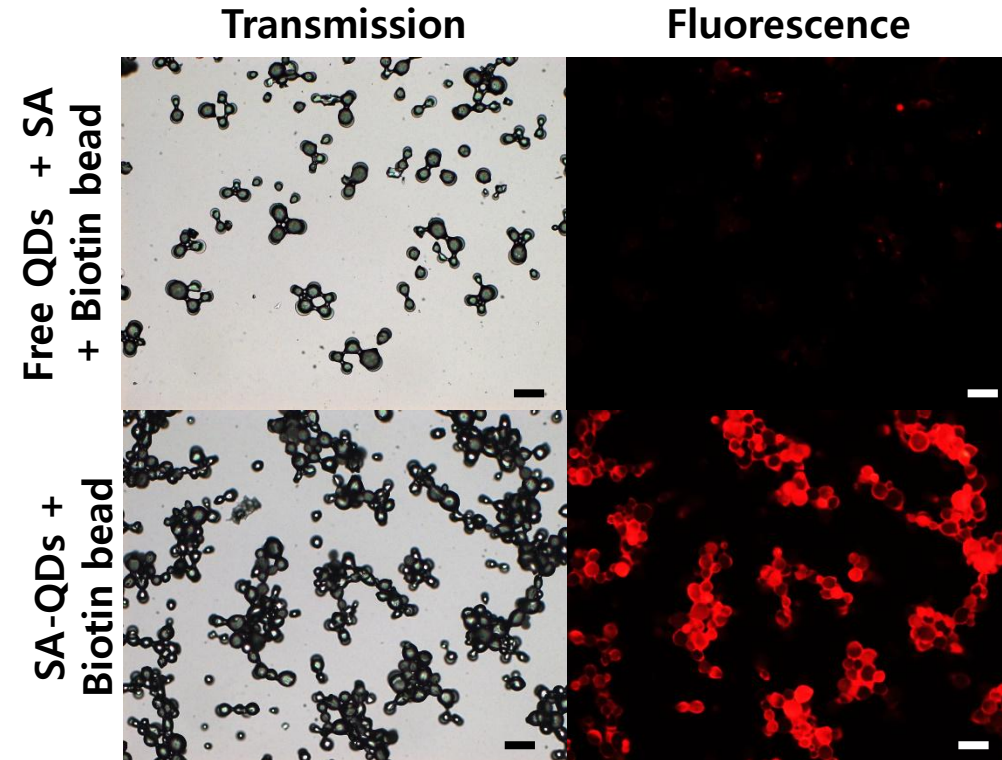
Research Experience: Nanomedicine

Polystyrene Bead assay



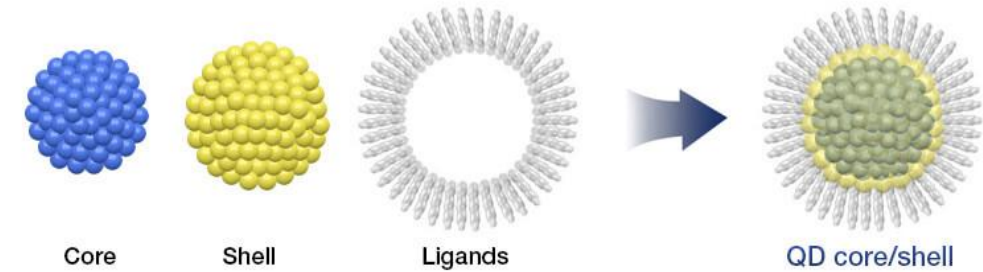
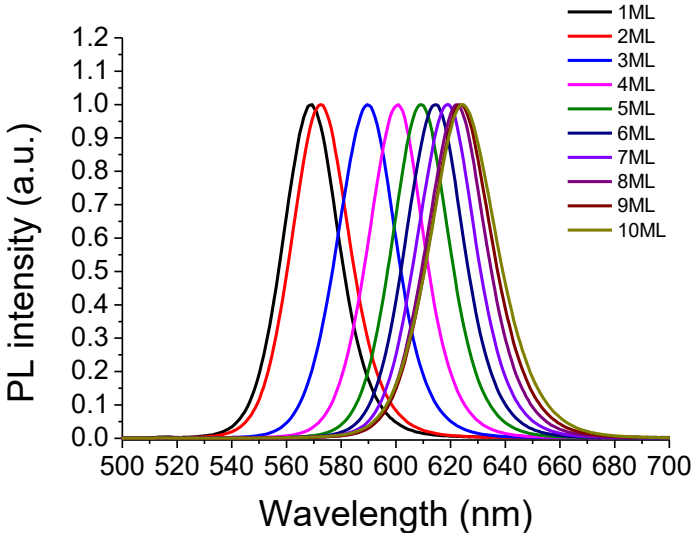
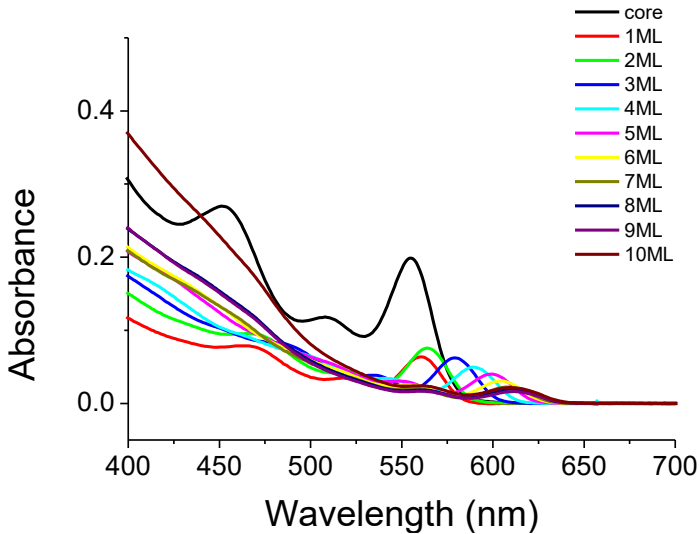
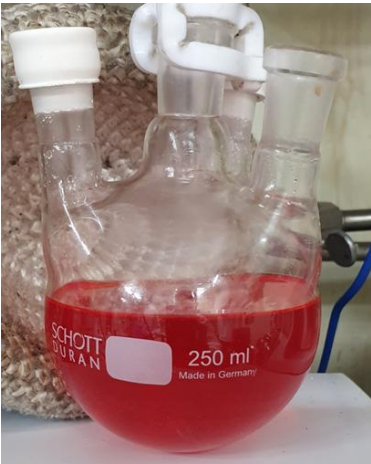
QD channel : (ex) 350/50 (em) 650 LP
Light integration time : 50 ms at fluorescence mode
Scale bar: 100 μ m

Biotinylated bead assay

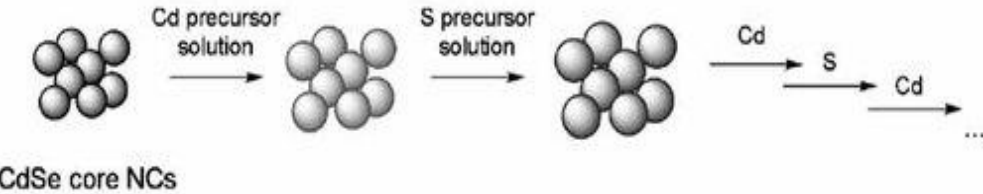


QD channel : (ex) 350/50 (em) 650 LP
Light integration time : 100 ms at fluorescence mode
Scale bar: 100 μ m

Appendix: Synthesis of CdSe/CdS/ZnS QD



SILAR technique



	Core	1 ML	2 ML	3 ML	4 ML	5 ML	6 ML	7 ML	8 ML	9 ML
1 st excitonic peak (nm)	555	560	579	589	599	604	608	611	612	609
PL peak (nm)	-	569.4	589.4	600.9	609.4	614.8	619.4	622.4	623.2	624.7
QY (%)	-	38.6	88.4	95.3	97.2	96.5	93.1	87.0	83.4	73.3

